

Moein Razavi, Ph.D.

Senior Data Scientist · Machine Learning & Statistical Modeling · Risk Analytics

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linkedin.com/in/moein-razavi · github.com/moeinrazavi · Google Scholar (470+ citations)

Professional Summary

Ph.D. Data Scientist & ML Leader with **5+ years** building, validating, and deploying production statistical and machine learning models at scale. Published researcher (**470+ citations**, 16 papers) with deep expertise in **classification, time-series forecasting, clustering, deep learning**, and model validation (ROC/AUC, confusion matrices, backtesting). Proven ability to translate complex analytical work into **tangible business outcomes** for cross-functional stakeholders. Hands-on with **Python, PySpark, SQL, AWS, H2O**, and cloud-native data platforms processing **10+ TB** datasets. Passionate about **challenging production models**, identifying failure modes, and driving data-driven decision-making in risk management.

Technical Skills

Statistical Modeling & ML: Classification (Logistic Regression, XGBoost, Random Forest, SVM) · Clustering (K-Means, DBSCAN, Hierarchical) · Time-Series Forecasting (TFT, ARIMA, Prophet, Autoformer) · Deep Learning (CNN, LSTM, Transformers) · Sentiment Analysis · NLP · Model Validation (ROC/AUC, Confusion Matrix, Precision/Recall, Backtesting, Cross-Validation)

Languages & Tools: Python (Expert) · R · SQL · Scala · Java · PySpark · pandas · NumPy · SciPy · scikit-learn · XGBoost · PyTorch · TensorFlow · H2O · Hugging Face · Optuna · SHAP/LIME

Cloud & Big Data: AWS (S3, EC2, SageMaker, Lambda, Redshift) · GCP (BigQuery, Vertex AI, Cloud Run, Dataproc) · Azure · Spark · Hadoop · Airflow · Conda · Docker · Kubernetes · Terraform

Data & Databases: Relational Databases (PostgreSQL, MySQL, BigQuery, Redshift) · MongoDB · ETL/ELT Pipelines · Data Warehousing · Query Optimization · Looker · Dash · Streamlit

Leadership & Communication: Cross-Functional Collaboration · Stakeholder Presentations · Talent Development · Mentoring · Agile/Scrum · Technical Documentation · Research Publication

Professional Experience

Senior Data Scientist & ML Engineer

Aug 2023 – Present

Ford Motor Company Detroit, MI

- Led development of **challenger models** using XGBoost, gradient-boosted trees, and deep learning to benchmark against production forecasting systems, identifying model drift and improving prediction accuracy by **10%** across EV energy analytics
- Built **XGBoost regression models** for EV energy bill prediction by engineering features from vehicle telematics, utility rates, parking behavior, and home energy consumption, achieving **$R^2 = 0.85$** and validating via ROC/AUC and backtesting
- Trained **transformer-based time-series forecasting** models (TFT, Autoformer, Informer) with Optuna hyperparameter optimization, improving forecast precision by **10%** and reducing training time by **40%** via distributed GPU computing
- Engineered **large-scale data pipelines** processing **10+ TB** of EV telemetry using **PySpark**, BigQuery, and Airflow DAGs; built Looker Studio dashboards automating KPI reporting for **500+ stakeholders**
- Architected **multi-agent AI system** with LangChain, RAG, and vector databases (FAISS, ChromaDB) serving analytics across **10 US states**, cutting query response latency by **35%** for cross-functional teams of data scientists, risk experts, and product managers
- Deployed production **MLOps infrastructure** on GCP (Cloud Run, Vertex AI) with CI/CD (Tekton), **Conda** environments, and Terraform IaC, achieving **99.9%** uptime across multi-environment deployments

Jun 2023 – Sep 2024

AI/ML Researcher & Software Engineer

CereVu Medical, Inc. San Francisco Bay Area, CA

- Deployed **deep learning classification models** (LSTM, CNN) on wearable medical devices, improving patient monitoring accuracy by **15%** across 400+ patients in 5 clinical studies; validated with confusion matrices, ROC curves, and cross-validation
- Delivered scalable **Python microservices** (Flask, FastAPI) with real-time data streaming, handling **10K+ concurrent connections** and reducing latency by **40%**
- Built **React/Next.js analytics dashboard** with MongoDB, interactive visualization, and role-based access, streamlining clinician decision-making workflows by **25%**

Sep 2021 – May 2024

Research Assistant & ML Engineer

Texas A&M University College Station, TX

- Designed a **dual-branch transformer architecture** for wearable stress classification achieving **AUC 0.993** and **F1 0.965** in 5-fold group cross-validation; engineered 18-feature HRV extraction pipeline with Mixup/CutMix augmentation and focal loss
- Published **comprehensive scoping review** of 200+ studies on ML/DL for stress detection (**79 citations**), synthesizing best practices for data preprocessing, feature engineering, and model selection
- Co-authored **spatial ML study** combining **unsupervised clustering and supervised classification** to identify county-level COVID-19 risk factors across the US, integrating demographic, socioeconomic, and spatial features (**9 citations**)
- Deployed **computer vision system** for COVID-19 safety compliance using deep learning object detection, achieving **98% accuracy** monitoring **50K+ daily interactions** (**76 citations**)
- Created **OpenSync**—open-source platform for real-time synchronization of multi-modal sensor data via Web-Socket pipelines, adopted by **3 research labs** (**31 citations**)

Education

Ph.D., Industrial Engineering (Data Science Concentration) | GPA: 4.0/4.0 | Dean's

Aug 2021 – Dec 2023

List | Research Fellowship

Texas A&M University, College Station, TX

Published **16 peer-reviewed papers** with **470+ citations**. Dissertation on ML/DL for stress detection and health monitoring.

Coursework: Machine Learning, Statistical Modeling, Data Mining, Optimization, Healthcare Analytics

Aug 2019 – May 2021

M.S., Computer Science | GPA: 4.0/4.0 | Dean's List | Research Fellowship

Texas A&M University, College Station, TX

Completed **12+ graduate courses** with a perfect GPA across Deep Learning, NLP, Computer Vision, and Database Systems.

Certifications

Deep Learning Specialization – DeepLearning.AI · **Machine Learning Specialization** – Stanford · **NLP Specialization** – DeepLearning.AI · **GenAI with LLMs** – AWS · **Bayesian Statistics** – Coursera

Selected Publications

Razavi, M., et al. "ML & Deep Learning for Stress Detection: Scoping Review." *JMIR Mental Health*, 2024. [**79 citations**]

Razavi, M., et al. "Automatic Monitoring for Physical Distance & Mask Compliance in COVID-19." *SN Computer Science*, 2022. [**76 citations**]

Razavi, M., et al. "OpenSync: Synchronizing Measures in Neuroscience." *J. Neuroscience Methods*, 2022. [**31 citations**]

Ziyadidegan, S., **Razavi, M.**, et al. "Factors Affecting COVID-19 Risk: Unsupervised & Supervised Learning." *SERR*, 2022. [**9 citations**]